Two Formulations of Robustness in Stochastic Optimization

Stability to Perturbations and Distributionally Robust Optimization

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Decisions under an approximate model

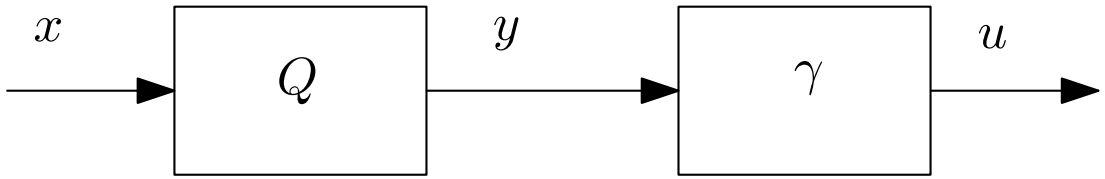
A decision depends on an uncertain state X, such as demand, a return, or a system disturbance.

- The true probability model is P.
- Data or an approximation give a model P_n.
- We choose a decision to minimize expected cost.

If $P_n \to P$, do our decisions become reliable?

For clarity, we first look at the static setting:

Static Single-Stage Bayesian Decision Problem



- State: $X \sim P$, where P is the prior distribution.
- Observation: $Y \sim Q(\cdot \mid X)$, where Q is the measurement channel.
- Action: $U = \gamma(Y)$, where $\gamma : \mathbb{Y} \to \mathbb{U}$ is a measurable policy.

Given a cost function $c : \mathbb{X} \times \mathbb{U} \to \mathbb{R}$, the expected cost is

$$J(P, Q, \gamma) = \int_{\mathbb{X}\times\mathbb{Y}} c(x, \gamma(y))\, Q(dy \mid x)\, P(dx).$$

Our objective is to choose a policy minimizing this cost:

$$J^*(P, Q) = \inf_{\gamma\in\Gamma} J(P, Q, \gamma).$$

If c is continuous in u, $\mathbb{U}$ compact, an optimal policy γ^* exists.

Related work: Kara and Yüksel (2019) [40]

Two formulations of robustness

Write $J(P,\gamma)$ for the expected cost of policy γ, and

$$J^*(P) = \inf_{\gamma} J(P,\gamma), \qquad \gamma^*_{P_n} \in \arg\min_{\gamma} J(P_n,\gamma).$$

Stability to Perturbations (RSP)

$$P_n \to P \quad \Longrightarrow \quad J(P,\gamma^*_{P_n}) \to J^*(P).$$

Distributionally Robust Optimization (DRO)

$$V_r(P) := \inf_{\gamma} \sup_{\bar{P}\in\mathcal{B}_r^{\rho}(P)} J(\bar{P},\gamma) \longrightarrow J^*(P) \quad (r \downarrow 0).$$

Here $\mathcal{B}_r^{\rho}(P) = \{\bar{P} : \rho(\bar{P},P) \le r\}$, with ρ a metric on $\mathcal{P}(\mathbb{X})$.

Related work: Kara and Yüksel (2019) [40]; Blanchet et al. (2025) [17]

The two properties need not imply each other

Theorem 2.3: Non-equivalence of DRO and RSP

The two robustness formulations are not comparable in general:

$$\text{RSP} \not\Longrightarrow \text{DRO}, \quad \text{and} \quad \text{DRO} \not\Longrightarrow \text{RSP}.$$

- RSP tests policies optimized for individual approximate models.
- DRO optimizes a policy against a whole neighborhood.

The paper gives counterexamples in both static and dynamic problems.

A nearby model chooses the wrong action

No informative observation. Choose $u \in [0,1]$, with

$$\mathbb{X} = \{0\} \cup \{1/n : n \geq 1\}, \qquad P = \delta_0, \quad P_n = \delta_{1/n} \to P.$$

$$c(0,u) = u^2, \qquad c(1/n,u) = \frac{1}{n}(1-u)^2.$$

Model	Optimal action	Optimal cost
True model P	0	0
Approximate model P_n	1	0

$$J^*(P_n) = J^*(P) = 0, \qquad J(P, u^*_{P_n}) = 1 \neq 0.$$

DRO still succeeds in this example

Choose the protective action $u = 0$. Then

$$c(0,0) = 0, \qquad c(1/n, 0) = 1/n.$$

Thus $c(x, 0) = x$ is bounded and continuous on $\mathbb{X}$. For a metric ρ inducing weak convergence,

$$0 \le V_r(P) \le \sup_{\bar{P} \in \mathcal{B}_r^\rho(P)} \int x \, \bar{P}(dx) \longrightarrow 0 = J^*(P).$$

DRO holds, while RSP fails.

DRO and Continuity of the Optimal Value

Let ρ metrize weak convergence, and define

$$V_n(P) := \inf_{\gamma \in \Gamma} \sup_{\bar{P} \in \mathcal{B}^{\rho}_{1/n}(P)} J(\bar{P}, \gamma).$$

Lemma 2.7: DRO implies upper semicontinuity

If $V_n(P) \to J^*(P)$, then

$$P_n \to P \quad \Longrightarrow \quad \limsup_{n \to \infty} J^*(P_n) \leq J^*(P).$$

Theorem 2.8: DRO implies value convergence

If, additionally, J^* is lower semicontinuous at P, then

$$P_n \to P \quad \Longrightarrow \quad J^*(P_n) \to J^*(P).$$

Conditions for Equivalence of DRO and RSP

Assumption

Let ρ metrize weak convergence.

(i) Suppose J^* is lower semicontinuous at P, and for every $P_n \in \mathcal{B}^{\rho}_{1/n}(P)$ and optimal policy $\gamma^*_{P_n}$,

$$\left|J(P_n, \gamma^*_{P_n}) - J(P, \gamma^*_{P_n})\right| \longrightarrow 0.$$

(ii) $Q(dy|x) = f(x,y)\phi(dy)$, Q continuous in total variation, and c continuous and bounded.

Theorem 2.9

Under (i),

$$\text{DRO} \Longrightarrow \text{RSP}.$$

If, additionally, (ii) holds, then

$$\text{DRO} \Longleftrightarrow \text{RSP}.$$

Extension to dynamic control

A transition kernel describes the next state:

$$X_{t+1} \sim \mathcal{T}(\cdot \mid X_t, U_t).$$

A policy chooses actions over time to minimize discounted cost:

$$J_\beta(P, \mathcal{T}, \gamma) = E_P^{\mathcal{T},\gamma}\left[\sum_{t=0}^{\infty} \beta^t c(X_t, U_t)\right], \qquad 0 < \beta < 1.$$

- RSP: optimize for $\mathcal{T}_n$, then use the policy under $\mathcal{T}$.
- DRO: optimize against nearby transition kernels.

The prior and cost stay fixed. Only the dynamics vary. Non-equivalence results once again hold in the dynamic setting. We focus on positive results.

Dynamic Robustness under Weak Convergence

RSP is known to hold under mild conditions: If $\mathcal{T}_n \to \mathcal{T}$ continuously weakly. For DRO we state the following: let ρ be an integral probability metric that metrizes weak convergence. At each state x, consider

$$D_x^n \subseteq \left\{ q \in C(\mathbb{U}, \mathcal{P}(\mathbb{X})) : \sup_{u \in \mathbb{U}} \rho\big(q(\cdot|u), \mathcal{T}(\cdot|x,u)\big) \leq \frac{1}{n} \right\}.$$

Under DRO: Controller minimizes over Γ and nature maximizes over D_x^n.

Assumption 3.2 and the ambiguity-set conditions

- c is bounded and continuous on $\mathbb{X} \times \mathbb{U}$, and $\mathbb{U}$ is compact.
- $\mathcal{T}$ and the approximating kernels are weakly continuous.
- Each D_x^n is nonempty, compact, and convex, with a common equicontinuity condition in u.
- $x \mapsto D_x^n$ is upper semicontinuous.

Related work: Kara and Yüksel (2020) [41]; Saldi and Yüksel (2026) [56]

Dynamic Robustness under Total Variation

Now endow $\mathcal{P}(\mathbb{X})$ with total variation and take

$$D_x^n \subseteq \left\{ q \in C(\mathbb{U}, \mathcal{P}(\mathbb{X})) : \sup_{u \in \mathbb{U}} \|q(\cdot|u) - \mathcal{T}(\cdot|x,u)\|_{\mathrm{TV}} \leq \frac{1}{n} \right\}.$$

Assumption 3.3 and the ambiguity-set conditions

- $\mathbb{U}$ is compact; c is bounded and measurable, with $c(x,\cdot)$ continuous.
- $\mathcal{T}$ is TV-continuous; admissible sections $u \mapsto q(\cdot|u)$ are TV-continuous.
- The statewise classes are compact and convex, with common equicontinuity and a TV-relatively compact measure range.
- $x \mapsto D_x^n$ has a measurable graph.

Robust Iterations and the DRO Value

For $J_0^n \equiv 0$, where

$$J_{m+1}^n(x) = \inf_{\gamma(\cdot|x)\in\mathcal{P}(\mathbb{U})} \sup_{\mathcal{R}(\cdot|x,\cdot)\in D_x^n} \int_{\mathbb{U}} \left(c(x,u) + \beta \int_{\mathbb{X}} J_m^n(x')\mathcal{R}(dx'|x,u) \right) \gamma(du|x).$$

Propositions 3.10 and 3.15: Contraction

Under the respective hypotheses,

$$\|J_{m+1}^n - \bar{J}_{m+1}^n\|_\infty \leq \beta \|J_m^n - \bar{J}_m^n\|_\infty, \qquad \|J_m^n - J_\infty^n\|_\infty \longrightarrow 0.$$

The fixed point belongs to $C_b(\mathbb{X})$ in the weak case and $B_b(\mathbb{X})$ in the TV case.

Theorems 3.12 and 3.17: Value identification

For the corresponding global ambiguity class D^n,

$$J_\infty^n(x) = \inf_{\gamma\in\Gamma} \sup_{R\in D^n} J_\beta(x,R,\gamma).$$

DRO, Value Convergence, and RSP

Under the corresponding weak or TV hypotheses, let $R_n \in D^n$, and let $\gamma^*_{R_n}$ satisfy the discounted optimality equation at every state. DRO and RSP are equivalent in D^n_x.

Theorems 3.13 and 3.18: DRO convergence

For every initial state x,

$$\lim_{n\to\infty} \lim_{m\to\infty} J^n_m(x) = J^*_\beta(x, \mathcal{T}).$$

Theorems 3.14 and 3.19: Value convergence and RSP

For the admissible approximating kernels,

$$\underbrace{J^*_\beta(x, R_n) \longrightarrow J^*_\beta(x, \mathcal{T})}_{\text{Value convergence}},$$

$$\underbrace{J_\beta(x, \mathcal{T}, \gamma^*_{R_n}) \longrightarrow J^*_\beta(x, \mathcal{T})}_{\text{Robustness in Stability to Perturbations (RSP)}} \quad .$$

Backup: the continuity needed for dynamics

A useful shrinking constraint is

$$\sup_{x,u} \rho\big(R_n(\cdot \mid x,u), \mathcal{T}(\cdot \mid x,u)\big) \le 1/n.$$

If $\mathcal{T}$ is weakly continuous and $(x_n,u_n) \to (x,u)$, then

$$\begin{aligned}&\rho\big(R_n(\cdot \mid x_n,u_n), \mathcal{T}(\cdot \mid x,u)\big)\\ &\quad\le \frac{1}{n} + \rho\big(\mathcal{T}(\cdot \mid x_n,u_n), \mathcal{T}(\cdot \mid x,u)\big) \to 0.\end{aligned}$$

This gives continuous weak convergence.

Statewise compactness requires a common equicontinuity condition and appropriate compactness of the measure range. Identifying the global DRO value also requires admissible measurable pasting.